

2. Creating the cluster

Clusters

Edit Config

+ Create

Cluster0

Connect

View Monitoring

Browse Collections

...

FREE

Visualize Your Data

Build dashboards and charts, and embed them in your apps with MongoDB Charts.

Explore Charts

Dismiss

**Cluster0 is ready to use!**

Monitoring will be available soon...

VERSION	REGION	TYPE	BACKUPS	LINKED APP SERVICES	ATLAS SQL	ATLAS SEARCH
8.0.23	AWS / Mumbai (ap-south-1)	Replica Set - 3 nodes	Inactive	None Linked	Connect	Create Index

+ Add Tag

Connected to mongoshell

```
C:\Users\ADMIN>mongosh "mongodb+srv://cluster0.ypvynfo.mongodb.net/" --apiVersion 1 --username admin
Enter password: *****
Current Mongosh Log ID: 69fec90341f9c8f0097c2906
Connecting to:      mongodb+srv://<credentials>@cluster0.ypvynfo.mongodb.net/?appName=mongosh+2.7.0
Using MongoDB:      8.0.23 (API Version 1)
Using Mongosh:       2.7.0
mongosh 2.8.3 is available for download: https://www.mongodb.com/try/download/shell

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

Atlas atlas-wi47rr-shard-0 [primary] test>
```

Insert the data

```
Atlas atlas-wi47rr-shard-0 [primary] test> use ecommerce
switched to db ecommerce
Atlas atlas-wi47rr-shard-0 [primary] ecommerce> db.createCollection(orders)
ReferenceError: orders is not defined
Atlas atlas-wi47rr-shard-0 [primary] ecommerce> db.createCollection('orders')
{ ok: 1 }
Atlas atlas-wi47rr-shard-0 [primary] ecommerce> show collections
orders
Atlas atlas-wi47rr-shard-0 [primary] ecommerce> db.orders.insertMany([
|   { "order_id": "ORD001", "customer": "Alice", "city": "Mumbai",
|     "category": "Electronics", "product": "Laptop", "quantity": 1,
|     "price": 75000, "status": "delivered", "date": "2024-01-15" },
|   { "order_id": "ORD002", "customer": "Bob", "city": "Delhi",
|     "category": "Clothing", "product": "Jacket", "quantity": 3,
|     "price": 4500, "status": "pending", "date": "2024-02-20" },
|   { "order_id": "ORD003", "customer": "Carol", "city": "Mumbai",
|     "category": "Electronics", "product": "Phone", "quantity": 2,
|     "price": 25000, "status": "delivered", "date": "2024-03-10" },
|   { "order_id": "ORD004", "customer": "Dave", "city": "Pune",
|     "category": "Books", "product": "Big Data Guide", "quantity": 5,
|     "price": 800, "status": "shipped", "date": "2024-03-25" },
|   { "order_id": "ORD005", "customer": "Eve", "city": "Delhi",
|     "category": "Electronics", "product": "Tablet", "quantity": 1,
|     "price": 35000, "status": "delivered", "date": "2024-04-05" }
| ])
| {
|   acknowledged: true,
|   insertedIds: {
|     '0': ObjectId('69fec9e141f9c8f0097c2907'),
|     '1': ObjectId('69fec9e141f9c8f0097c2908'),
|     '2': ObjectId('69fec9e141f9c8f0097c2909'),
|     '3': ObjectId('69fec9e141f9c8f0097c290a'),
|     '4': ObjectId('69fec9e141f9c8f0097c290b')
|   }
| }
Atlas atlas-wi47rr-shard-0 [primary] ecommerce>
```

CRUD operations :

```
Atlas atlas-wi47rr-shard-0 [primary] ecommerce> db.orders.find({status:"delivered"},{_id:0})
[
  {
    order_id: 'ORD001',
    customer: 'Alice',
    city: 'Mumbai',
    category: 'Electronics',
    product: 'Laptop',
    quantity: 1,
    price: 75000,
    status: 'delivered',
    date: '2024-01-15'
  },
  {
    order_id: 'ORD003',
    customer: 'Carol',
    city: 'Mumbai',
    category: 'Electronics',
    product: 'Phone',
    quantity: 2,
    price: 25000,
    status: 'delivered',
    date: '2024-03-10'
  },
  {
    order_id: 'ORD005',
    customer: 'Eve',
    city: 'Delhi',
    category: 'Electronics',
    product: 'Tablet',
    quantity: 1,
    price: 35000,
    status: 'delivered',
    date: '2024-04-05'
  }
]
Atlas atlas-wi47rr-shard-0 [primary] ecommerce>
```

```
Atlas atlas-wi47rr-shard-0 [primary] ecommerce> db.orders.updateOne({order_id:'ORD002'},{$set:{status:'shipped'}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-wi47rr-shard-0 [primary] ecommerce>
```

```
Atlas atlas-wi47rr-shard-0 [primary] ecommerce> db.orders.insertOne({ "order_id": "ORD006", "customer": "Jay", "city": "Mumbai",
|   "category": "Electronics", "product": "Laptop", "quantity": 1,
|   "price": 90000, "status": "shipped", "date": "2024-05-06" })
{
  acknowledged: true,
  insertedId: ObjectId('69fecc4e41f9c8f0097c290c')
}
Atlas atlas-wi47rr-shard-0 [primary] ecommerce>
```

```
Atlas atlas-wi47rr-shard-0 [primary] ecommerce> db.orders.deleteOne({order_id:'ORD004'})
{ acknowledged: true, deletedCount: 0 }
Atlas atlas-wi47rr-shard-0 [primary] ecommerce>
```

Aggregate pipeline:

```
Atlas atlas-wi47rr-shard-0 [primary] ecommerce> db.orders.aggregate([{$group: {_id: '$category', total_revenue: {$sum: {$multiply: ['$price', '$quantity']}}}}, {$sort: {total_revenue: -1}}])
[
  { _id: 'Electronics', total_revenue: 250000 },
  { _id: 'Clothing', total_revenue: 13500 },
  { _id: 'Books', total_revenue: 4000 }
]
Atlas atlas-wi47rr-shard-0 [primary] ecommerce>
```

```
Atlas atlas-wi47rr-shard-0 [primary] ecommerce> db.orders.aggregate([{$group: {_id: '$city', total_orders: {$sum: 1}, avg_price: {$avg: '$price'}}}])
[
  { _id: 'Mumbai', total_orders: 3, avg_price: 63333.333333333336 },
  { _id: 'Delhi', total_orders: 2, avg_price: 19750 },
  { _id: 'Pune', total_orders: 1, avg_price: 800 }
]
Atlas atlas-wi47rr-shard-0 [primary] ecommerce>
```

```
Atlas atlas-wi47rr-shard-0 [primary] ecommerce> db.orders.aggregate([{$group: {_id: '$status', count: {$sum: 1}}}, {$match: {count: {$gt: 1}}}])
[ { _id: 'delivered', count: 3 }, { _id: 'shipped', count: 3 } ]
Atlas atlas-wi47rr-shard-0 [primary] ecommerce>
```

1. Create an EMR cluster

Amazon EMR > EMR on EC2: Clusters > my_practiac_lcluster_43

Your cluster 'my_practiac_lcluster_43' has been successfully created.

my_practiac_lcluster_43 Updated 7 minutes ago [Terminate](#) [Clone in AWS CLI](#) [Clone](#)

This EMR release reaches End of Support on Aug-01-2026 and will no longer be eligible for technical support. AWS strongly recommends that you run your workloads on the latest Amazon EMR release to receive security-critical updates and fixes. You can also use new Spark upgrade agent to modernize your existing applications on version 5.40 or higher to latest EMR version. To learn more, see [EMR Standard Support policy](#) and [Spark Upgrades](#).

▼ Summary

Cluster info

Cluster ID
j-2OUO6P68CH4RN

Cluster ARN
arn:aws:elasticmapreduce:us-east-1:471112753524:cluster/j-2OUO6P68CH4RN

Cluster configuration
Instance groups

Capacity
1 Primary | 1 Core | 0 Task

Applications

Amazon EMR version
emr-5.20.1

Installed applications
Ganglia 3.7.2, Hadoop 2.8.5, Hive 2.3.4, Hue 4.3.0, Tez 0.9.1

Cluster management

Log destination in Amazon S3
Logging not configured

Primary node public DNS
ec2-100-27-2-158.compute-1.amazonaws.com
[Connect to the Primary node using SSH](#)
[Connect to the Primary node using SSM](#)

Status and time

Status
Waiting

Creation time
9 May 2026 11:02 (UTC+05:30)

Elapsed time
48 minutes, 29 seconds

Properties

Bootstrap actions

Instances (hardware)

Steps

Applications

Configurations

Monitoring

Events

Tags (0)

Connected to EMR cluster via MobaXterm

```

• MobaXterm Personal Edition v26.3 •
(SSH client, X server and network tools)

➤ SSH session to hadoop@ec2-100-27-2-158.compute-1.amazonaws.com
  • Direct SSH      : ✓
  • SSH compression : ✓
  • SSH-browser     : ✓
  • X11-forwarding  : ✗ (disabled or not supported by server)

➤ For more info, ctrl+click on help or visit our website.

```

Last login: Sat May 9 06:24:15 2026

```

 _ | _ | )
 _ | ( _ /   Amazon Linux AMI

```

https://aws.amazon.com/amazon-linux-ami/2018_03-release-notes/

54 package(s) needed for security, out of 80 available
Run "sudo yum update" to apply all updates.

```

EEEEEEEEEEEEEEEEEE MMMMMMMM                MMMMMMMM RRRRRRRRRRRRRR
E:::EEEEEEEE:::E M:::MM          M:::MM R:::R:::R:::R
E:::EEEEEEEE:::E M:::MM          M:::MM R:::RARRRR:::R
 E:::E         EEEEE M:::MM          M:::MM RR:::R    R:::R
E:::E         EEEEE M:::MM          M:::MM RR:::R    R:::R
E:::EEEEEEEEEE M:::MM M:::MM M:::MM M:::MM R:::RARRRR:::R
E:::EEEEEEEEEE M:::MM M:::MM M:::MM M:::MM R:::RARRRR:::R
E:::EEEEEEEEEE M:::MM M:::MM M:::MM M:::MM R:::RARRRR:::R
E:::E         M:::MM M:::MM M:::MM M:::MM R:::R    R:::R
E:::E         EEEEE M:::MM M:::MM M:::MM R:::R    R:::R
EE:::EEEEEEEE:::E M:::MM          M:::MM R:::R    R:::R
E:::EEEEEEEE:::E M:::MM          M:::MM RR:::R    R:::R
EEEEEEEEEEEEEEEEEE MMMMMMMM                MMMMMMMM RRRRRRR    RRRRRR

```

[hadoop@ip-172-31-63-182 ~]\$ █

Connected to hue

The screenshot shows the Apache Hue web interface. At the top, there's a navigation bar with the Hue logo, a "Query" dropdown menu, a search bar labeled "Search saved documents...", and links for "Jobs", "Refresh", and "hadoop". Below this is a sidebar on the left containing icons for various tools and a "Tables" section showing "(0) + ↻" and "No entries found". The main area is titled "File Browser" and contains a search bar "Search for file name", action buttons "Actions" and "Delete forever", and "Upload" and "New" buttons. The breadcrumb path is "Home / user / hadoop". A table lists files and directories:

☐	Name	Size	User	Group	Permissions	Date
☒	j		hdfs	hadoop	drwxr-xr-x	May 08, 2026 10:37 PM
☐	.		hadoop	hadoop	drwxrwxrwx	May 08, 2026 10:37 PM

At the bottom, it says "Show 45 of 0 items" and "Page 1 of 1" with pagination controls.

Created external table and loaded csv file into table:

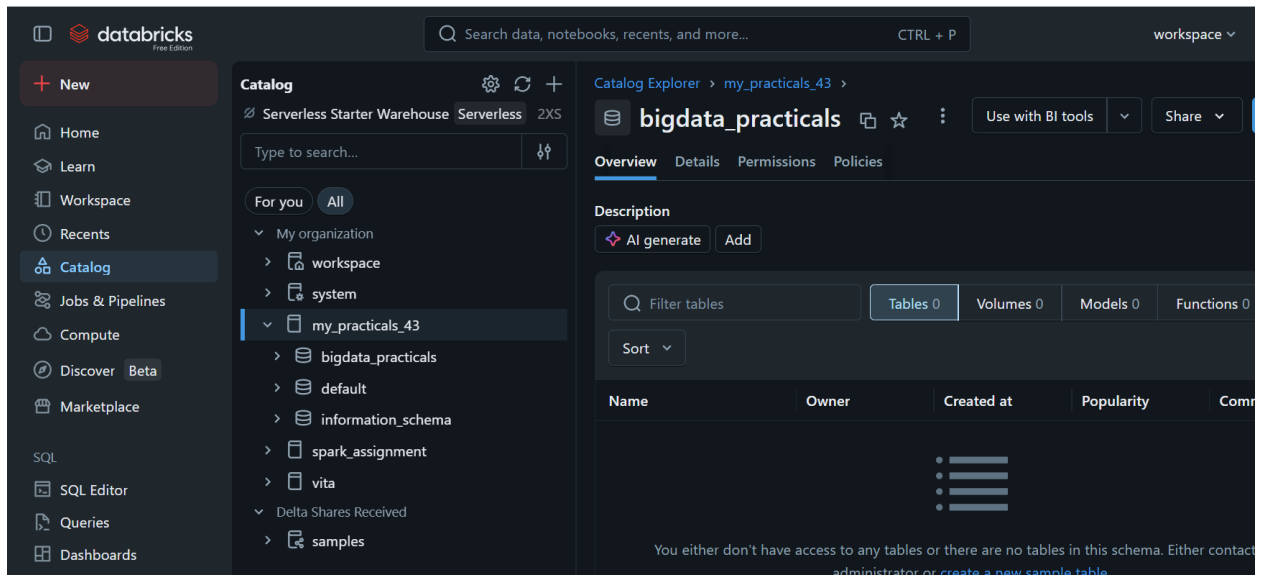
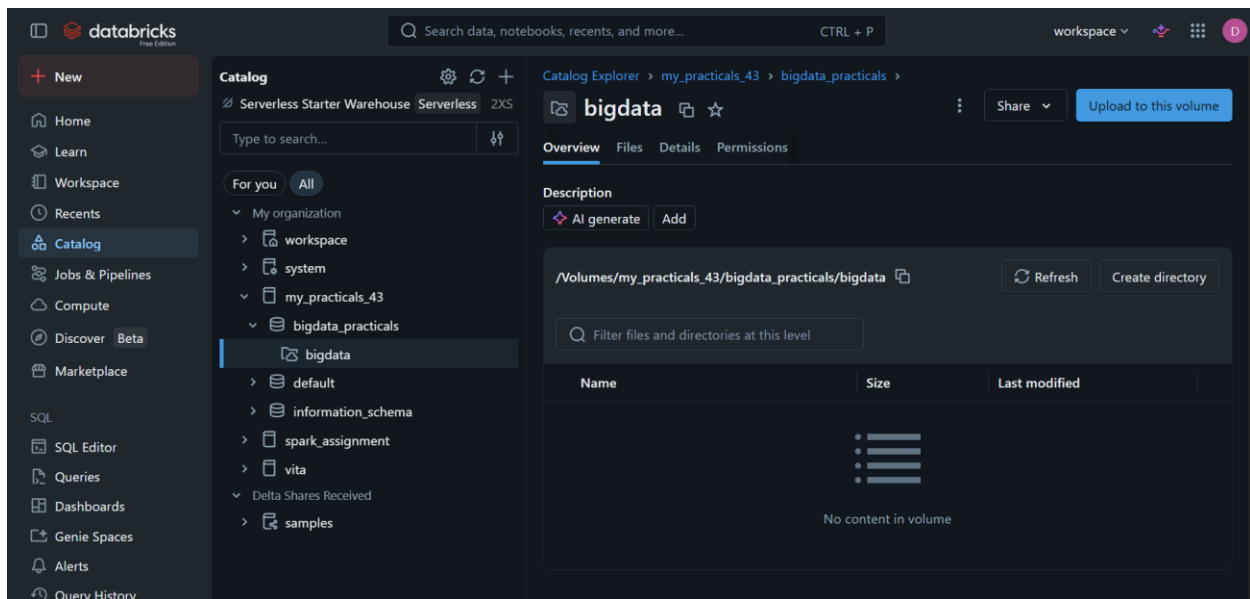
```
hive> CREATE EXTERNAL TABLE crimes_raw (
>   crime_id BIGINT,
>   case_number STRING,
>   crime_date STRING,
>   block STRING,
>   iucr STRING,
>   primary_type STRING,
>   description STRING,
>   location_description STRING,
>   arrest BOOLEAN,
>   domestic BOOLEAN,
>   beat INT,
>   district INT,
>   ward INT,
>   community_area INT,
>   fbi_code STRING,
>   x_coordinate BIGINT,
>   y_coordinate BIGINT,
>   year INT,
>   updated_on STRING,
>   latitude DOUBLE,
>   longitude DOUBLE,
>   location STRING
> )
> ROW FORMAT DELIMITED
> FIELDS TERMINATED BY ','
> STORED AS TEXTFILE;
OK
Time taken: 0.831 seconds
hive> load data inpath '/user/hadoop/Chicago_Crimes.csv' into table crimes_raw;
Loading data to table crime_db.crimes_raw
OK
Time taken: 1.418 seconds
hive> 
```

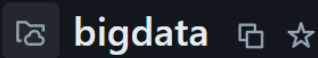
3.

The screenshot shows the Databricks Catalog Explorer interface. The left sidebar contains navigation options: Home, Learn, Workspace, Recents, Catalog (selected), Jobs & Pipelines, Compute, Discover (Beta), Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie Spaces, Alerts, and Query History. The main panel displays the 'my_practicals_43' catalog. The 'Catalog Explorer' header shows the catalog name, a search bar, and a 'Create schema' button. The 'Overview' tab is active, showing a description with an 'AI generate' button and an 'Add' button. Below this is a table of schemas:

Name	Owner	Created at
default	dandekarviraj1999@gmail.co...	May 09, 2026, 12:05 PM
information_schema	System user	May 09, 2026, 12:05 PM

Below the table, the 'About this catalog' section shows the owner as 'dandekarviraj1999@gmail.com'. The 'Tags' section has an 'Add tags' button.





Share ▾

Upload to this volume

Overview Files Details Permissions

Description

AI generate Add

/Volumes/my_practicals_43/bigdata_practicals/bigdata

Refresh

Create directory

Filter files and directories at this level

Name	Size	Last modified
airports.csv	12.03 MB	13 seconds ago
runways.csv	3.76 MB	18 seconds ago

About this volume

Owner dandekarviraj1999@gmail.com

26044525043_Viraj_Dandekar

File Edit View Run Help Python Last edit was now Run all Serverless Schedule Share

2 minutes ago (<1s) 1

```
import pyspark
```

Just now (3s) 2 Python

```
df_airports = spark.read.csv('/Volumes/my_practicals_43/bigdata_practicals/bigdata/airports.csv', header=True, inferSchema=True)
df_runways = spark.read.csv('/Volumes/my_practicals_43/bigdata_practicals/bigdata/runways.csv', header=True, inferSchema=True)
```

df_airports: pyspark.sql.connect.dataframe.DataFrame = [id: integer, ident: string ... 17 more fields]
df_runways: pyspark.sql.connect.dataframe.DataFrame = [id: integer, airport_ref: integer ... 18 more fields]

```
hive> CREATE TABLE crimes_partitioned (
>   crime_id BIGINT,
>   case_number STRING,
>   crime_date STRING,
>   block STRING,
>   iucr STRING,
>   primary_type STRING,
>   description STRING,
>   location_description STRING,
>   arrest BOOLEAN,
>   domestic BOOLEAN,
>   beat INT,
>   district INT,
>   ward INT,
>   community_area INT,
>   fbi_code STRING,
>   x_coordinate BIGINT,
>   y_coordinate BIGINT,
>   updated_on STRING,
>   latitude DOUBLE,
>   longitude DOUBLE,
>   location STRING
> )
> PARTITIONED BY (year INT)
> CLUSTERED BY (district)
> INTO 4 BUCKETS
> ROW FORMAT DELIMITED
> FIELDS TERMINATED BY ','
> STORED AS TEXTFILE;
```

OK

Time taken: 0.089 seconds

hive> SET hive.enforce.bucketing = true;

▶ ✓ 1 minute ago (<1s)

4

```
df_runways.printSchema()
```

```
|-- id: integer (nullable = true)
|-- airport_ref: integer (nullable = true)
|-- airport_ident: string (nullable = true)
|-- length_ft: integer (nullable = true)
|-- width_ft: integer (nullable = true)
|-- surface: string (nullable = true)
|-- lighted: integer (nullable = true)
|-- closed: integer (nullable = true)
|-- le_ident: string (nullable = true)
|-- le_latitude_deg: double (nullable = true)
|-- le_longitude_deg: double (nullable = true)
|-- le_elevation_ft: integer (nullable = true)
|-- le_heading_degT: double (nullable = true)
|-- le_displaced_threshold_ft: integer (nullable = true)
|-- he_ident: string (nullable = true)
```